

Лабораторная работа № 3. Измерение и тестирование пропускной способности сети. Воспроизводимый эксперимент

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Цель работы

Основной целью работы является знакомство с инструментом для измерения пропускной способности сети в режиме реального времени — iPerf3, а также получение навыков проведения воспроизводимого эксперимента по измерению пропускной способности моделируемой сети в среде Mininet.

Задание

1. Воспроизвести посредством API Mininet эксперименты по измерению пропускной способности с помощью iPerf3.
2. Построить графики по проведённому эксперименту.

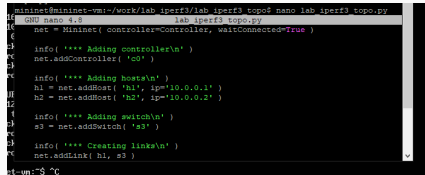
Выполнение лабораторной работы

Воспроизведение экспериментов посредством API Mininet

```
mininet@mininet-vm:~$ cd ~/work/lab_iperf3
mininet@mininet-vm:~/work/lab_iperf3$ mkdir lab_iperf3_topo
mininet@mininet-vm:~/work/lab_iperf3$ cd ~/work/lab_iperf3/lab_iperf3_topo
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ cp ~/mininet/examples/empt
ynet.py ~/work/lab_iperf3/lab_iperf3_topo
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ mv emptynet.py lab_iperf3_
topo.py
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$
```

Рис. 1: Создание подкаталога и копирование в него файла

Воспроизведение экспериментов посредством API Mininet



```
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ nano lab_iperf3_topo.py
GNU nano 4.8 lab_iperf3_topo.py
16 net = Mininet( controller=Controller, waitConnected=True )
17
18 info( '*** Adding controller\n' )
19 net.addController( 'c0' )
20
21 info( '*** Adding hosts\n' )
22 h1 = net.addHost( 'h1', ip='10.0.0.1' )
23 h2 = net.addHost( 'h2', ip='10.0.0.2' )
24
25 info( '*** Adding switch\n' )
26 s3 = net.addSwitch( 's3' )
27
28 info( '*** Creating links\n' )
29 net.addLink( h1, s3 )
30
31 mininet>
```

Рис. 2: Содержание скрипта lab_iperf3_topo.py

Воспроизведение экспериментов посредством API Mininet

```
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ sudo python lab_iperf3_topo.py
** Adding controller
** Adding hosts
** Adding switch
** Creating links
** Starting network
** Configuring hosts
h1 h2
** Starting controller
h0
** Starting 1 switches
h3 ...
** Waiting for switches to connect
h3
** Running CLI
** Starting CLI:
mininet>
```

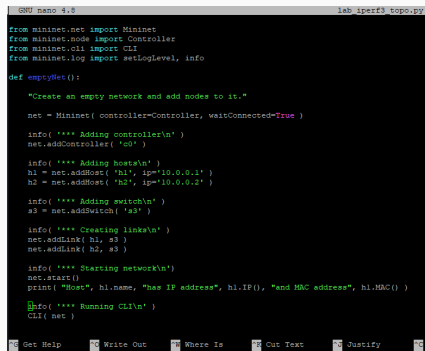
Рис. 3: Запуск скрипта создания топологии

Воспроизведение экспериментов посредством API Mininet

```
mininet> net
h1 h1-eth0:s3-eth1
h2 h2-eth0:s3-eth2
s3 lo: s3-eth1:h1-eth0 s3-eth2:h2-eth0
c0
mininet> links
h1-eth0<->s3-eth1 (OK OK)
h2-eth0<->s3-eth2 (OK OK)
mininet> dump
<Host h1: h1-eth0:10.0.0.1 pid=765>
<Host h2: h2-eth0:10.0.0.2 pid=768>
<OVSSwitch s3: lo:127.0.0.1,s3-eth1:None,s3-eth2:None pid=773>
<Controller c0: 127.0.0.1:6653 pid=758>
mininet> exit
*** Stopping network*** Stopping 1 controllers
c0
*** Stopping 2 links
..
*** Stopping 1 switches
s3
*** Stopping 2 hosts
h1 h2
*** Done
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$
```

Рис. 4: Просмотр элементов топологии

Воспроизведение экспериментов посредством API Mininet



The image shows a terminal window with a dark background and light-colored text. The title bar at the top reads "GNU nano 4.8" on the left and "lab_iperf3_topo.py" on the right. The main area contains Python code for setting up a Mininet network. The code includes imports for Mininet, Controller, CLI, and logging. A function named `emptyNet()` is defined, which creates a Mininet object, adds a controller, two hosts, and a switch, and then starts the network. At the bottom of the terminal, there is a status bar with various icons and labels like "Get Help", "Write Out", "Where Is", "Cut Text", "Justify", etc.

```
GNU nano 4.8 lab_iperf3_topo.py
from mininet.net import Mininet
from mininet.node import Controller
from mininet.cli import CLI
from mininet.log import setLogLevel, info

def emptyNet():
    "Create an empty network and add nodes to it."

    net = Mininet( controller=Controller, waitConnected=True )

    info( '*** Adding controller\n' )
    net.addController( 'c0' )

    info( '*** Adding hosts\n' )
    h1 = net.addHost( 'h1', ip='10.0.0.1' )
    h2 = net.addHost( 'h2', ip='10.0.0.2' )

    info( '*** Adding switch\n' )
    s3 = net.addSwitch( 's3' )

    info( '*** Creating links\n' )
    net.addLink( h1, s3 )
    net.addLink( h2, s3 )

    info( '*** Starting network\n' )
    net.start()
    print( "Host", h1.name, "has IP address", h1.IP(), "and MAC address", h1.MAC() )

    info( '*** Running CLI\n' )
    CLI( net )
```

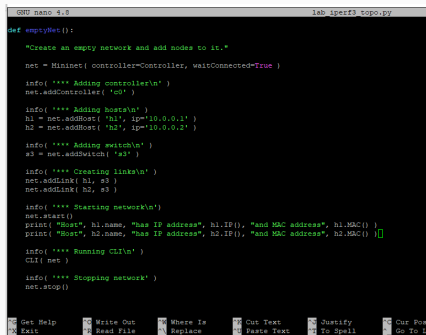
Рис. 5: Внесение изменения в скрипт `lab_iperf3_topo.py`

Воспроизведение экспериментов посредством API Mininet

```
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ nano lab_iperf3_topo.py
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ sudo python lab_iperf3_topo.py
*** Adding controller
*** Adding hosts
*** Adding switch
*** Creating links
*** Starting network
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 1 switches
s3 ...
*** Waiting for switches to connect
s3
Host h1 has IP address 10.0.0.1 and MAC address 6a:15:c9:20:1b:a8
*** Running CLI
*** Starting CLI:
mininet> █
```

Рис. 6: Успешная отработка скрипта

Воспроизведение экспериментов посредством API Mininet



```
GNU nano 4.0 lab_iperf3_topo.py
def emptyNet():
    """Create an empty network and add nodes to it."""
    net = Mininet( controller=Controller, waitConnected=True )

    info( """ Adding controller\n" )
    net.addController( 'c0' )

    info( """ Adding hosts\n" )
    h1 = net.addHost( 'h1', ip='10.0.0.1' )
    h2 = net.addHost( 'h2', ip='10.0.0.2' )

    info( """ Adding switch\n" )
    s3 = net.addSwitch( 's3' )

    info( """ Creating links\n" )
    net.addLink( h1, s3 )
    net.addLink( h2, s3 )

    info( """ Starting network\n" )
    net.start()
    print( "Host", h1.name, "has IP address", h1.IP(), "and MAC address", h1.MAC() )
    print( "Host", h2.name, "has IP address", h2.IP(), "and MAC address", h2.MAC() )

    info( """ Running CLI\n" )
    CLI( net )

    info( """ Stopping network\n" )
    net.stop()
```

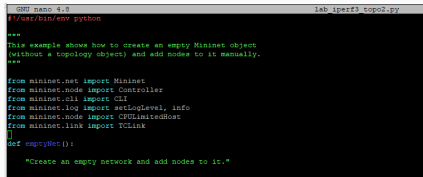
Рис. 7: Изменение скрипта для второго хоста

Воспроизведение экспериментов посредством API Mininet

```
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ nano lab_iperf3_topo.py
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ sudo python lab_iperf3_topo.py
*** Adding controller
*** Adding hosts
*** Adding switch
*** Creating links
*** Starting network
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 1 switches
s3 ...
*** Waiting for switches to connect
s3
Host h1 has IP address 10.0.0.1 and MAC address b6:7b:e5:bd:8c:9c
Host h2 has IP address 10.0.0.2 and MAC address f6:24:e1:24:71:2a
*** Running CLI
*** Starting CLI:
mininet>
```

Рис. 8: Успешная отработка скрипта с учетом второго хоста

Воспроизведение экспериментов посредством API Mininet



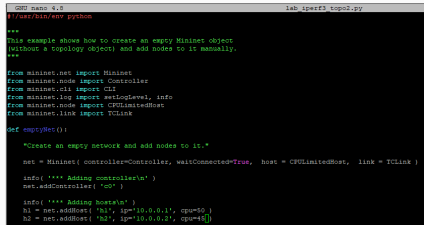
```
GNU nano 4.8 lab_iperf3_topo2.py
#!/usr/bin/env python

"""
This example shows how to create an empty Mininet object
(without a topology object) and add nodes to it manually.
"""

from mininet.net import Mininet
from mininet.node import Controller
from mininet.cli import CLI
from mininet.log import setLogLevel, info
from mininet.node import CPUimitedHost
from mininet.link import TCLink
]
def emptyNet():
    "Create an empty network and add nodes to it."
    net = Mininet(controller=Controller, switch=OpenFlowSwitch, link=TCLink)
```

Рис. 9: Добавление записей об импорте классов

Воспроизведение экспериментов посредством API Mininet



```
GNU nano 4.8 lab_iperf3_topo2.py
#!/usr/bin/env python

'''
This example shows how to create an empty Mininet object
(without a topology object) and add nodes to it manually.
'''

from mininet.net import Mininet
from mininet.node import Controller
from mininet.cli import CLI
from mininet.log import setLogLevel, info
from mininet.node import CPULimitedHost
from mininet.link import TCLink

def emptyNet():
    "Create an empty network and add nodes to it."

    net = Mininet( controller=Controller, waitConnected=True, host = CPULimitedHost, link = TCLink )

    info( '*** Adding controller\n' )
    net.addController( 'c0' )

    info( '*** Adding hosts\n' )
    h1 = net.addHost( 'h1', ip='10.0.0.1', cpu=10 )
    h2 = net.addHost( 'h2', ip='10.0.0.2', cpu=10 )
```

Рис. 10: Изменение строки описания сети в скрипте, функции задания параметров для обоих хостов

Воспроизведение экспериментов посредством API Mininet

```
s3 = Net.addSwitch('s3')  
info( '*** Creating link\n' )  
net.addLink( h1, s3, bw=10, delay='5ms', max_queue_size=1000, loss=10, use_htb=True )  
net.addLink( h2, s3 )  
  
info( '*** Starting network\n' )
```

Рис. 11: Изменение функции параметров соединения

Воспроизведение экспериментов посредством API Mininet

```
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ sudo python lab_iperf3_topo2.py
*** Adding controller
*** Adding hosts
*** Adding switch
*** Creating links
(10.00Mbit 5ms delay 10.00000% loss) (10.00Mbit 5ms delay 10.00000% loss) *** Starting network
*** Configuring hosts
h1 (cfs 5000000/1000000us) h2 (cfs 4500000/1000000us)
*** Starting controller
c0
*** Starting 1 switches
s3 (10.00Mbit 5ms delay 10.00000% loss) ... (10.00Mbit 5ms delay 10.00000% loss)
*** Waiting for switches to connect
s3
Host h1 has IP address 10.0.0.1 and MAC address 1a:fb:9e:0e:dd:d42
Host h2 has IP address 10.0.0.2 and MAC address 22:01:d4:ea:3f:50
*** Running CLI
*** Starting CLI:
mininet> exit
*** Stopping network*** Stopping 1 controllers
c0
(cfs -1/1000000us) (cfs -1/1000000us) *** Stopping 2 links
...
*** Stopping 1 switches
s3
*** Stopping 2 hosts
h1 h2
*** Done
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ sudo python lab_iperf3_topo.py
*** Adding controller
*** Adding hosts
*** Adding switch
*** Creating links
*** Starting network
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 1 switches
s3 ...
*** Waiting for switches to connect
s3
Host h1 has IP address 10.0.0.1 and MAC address 42:5b:06:0c:7d:30
Host h2 has IP address 10.0.0.2 and MAC address 42:29:86:8a:57:30
*** Running CLI
*** Starting CLI:
mininet>
```

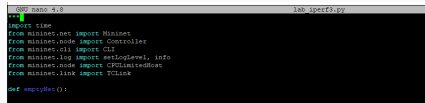
Рис. 12: Запуск обоих скриптов

Построение графика по проведенному эксперименту

```
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ cp lab_iperf3_topo.py lab_iperf3.py
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ mkdir -p ~/work/lab_iperf3/iperf3
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ mv ~/work/lab_iperf3/lab_iperf3_topo/lab_iperf3.py ~/work/lab_iperf3/iperf3
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ cd ~/work/lab_iperf3/iperf3
mininet@mininet-vm:~/work/lab_iperf3/iperf3$ ls -la
total 8
-rwxrwxr-x 1 mininet mininet 1347 Oct  7 17:02 lab_iperf3.py
```

Рис. 13: Создание копии скрипта lab_iperf3_topo2.py и помещение в подкаталог

Построение графика по проведенному эксперименту



```
GNU nano 4.0 lab_iperf3.py
***
import time
from mininet.net import Mininet
from mininet.node import Controller
from mininet.cli import CLI
from mininet.log import setLogLevel, info
from mininet.node import CPUimitedHost
from mininet.link import TCLink
def emptyNet():
```

Рис. 14: Добавление записи в начало скрипта lab_iperf3.py

Построение графика по проведенному эксперименту

```
s3 = net.addSwitch( 's3' )  
  
info( '*** Creating links\n' )  
net.addLink( h1, s3, bw=100, delay='75ms' )  
net.addLink( h2, s3, bw=100, delay='75ms' )
```

Рис. 15: Внесение изменений в скрипт для хостов

Построение графика по проведенному эксперименту

```
net.start()
print( "Host", h1.name, "has IP address", h1.IP(), "and MAC address", h1.MAC() )
print( "Host", h2.name, "has IP address", h2.IP(), "and MAC address", h2.MAC() )
info( '*** Starting network\n' )
info( '*** Traffic generation\n' )
h2.cmdPrint( 'iperf3 -s -D -l' )
time.sleep(10) # Wait 10 seconds for servers to start
h1.cmdPrint( 'iperf3 -c', h2.IP(), '-J > iperf_result.json' )
# info( '*** Running CLI\n' )
# CLI( net )
]
```

Рис. 16: Описание запуска на хостах и комментирование строк

Построение графика по проведенному эксперименту

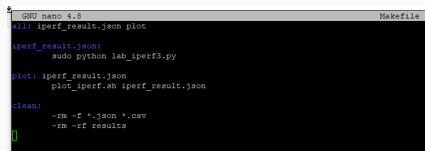
```
-overpass-X 1 mininet mininet 134' Oct 7 07:52 lab_iperf3.py
mininet@mininet-vm:~/work/lab_iperf3/iperf3$ nano lab_iperf3.py
mininet@mininet-vm:~/work/lab_iperf3/iperf3$ sudo python lab_iperf3.py
*** Adding controller
*** Adding hosts
*** Adding switch
*** Creating links
(100.00Mbit 75ms delay) (100.00Mbit 75ms delay) (100.00Mbit 75ms delay) (100.00Mbit 75ms delay) *** Starting network
*** Configuring hosts
h1 (cfe -1/10000bus) h2 (cfe -1/10000bus)
*** Starting controller
c0
*** Starting 1 switches
s1 (100.00Mbit 75ms delay) (100.00Mbit 75ms delay) ... (100.00Mbit 75ms delay) (100.00Mbit 75ms delay)
*** Waiting for switches to connect
s1
Host h1 has IP address 10.0.0.1 and MAC address 0a:c9:52:d3:52:d4
Host h2 has IP address 10.0.0.2 and MAC address 0a:rad:46:3f:d3:3c
*** Starting network
*** Traffic generation
*** h2 : ('iperf3' -s -D -i 1)
*** h1 : ('iperf3' -c, '10.0.0.2', '-J > iperf_result.json)
*** Stopping network*** Stopping 1 controllers
c0
*** Stopping 2 links
..
*** Stopping 1 switches
s1
*** Stopping 2 hosts
h1 h2
```

Рис. 17: Успешная отработка скрипта lab_iperf3.py

```
mininet@mininet-vm:~/work/lab_iperf3/iperf3$ plot_iperf.sh iperf_result.json  
mininet@mininet-vm:~/work/lab_iperf3/iperf3$ touch Makefile  
mininet@mininet-vm:~/work/lab_iperf3/iperf3$
```

Рис. 18: Построение графиков из JSON-файла и создание MAkefile

Построение графика по проведенному эксперименту



```
GNU nano 4.8 Makefile
all: iperf_result.json plot

iperf_result.json:
    sudo python lab_iperf3.py

plot: iperf_result.json
    plot_iperf.sh iperf_result.json

clean:
    -rm -rf *.json *.csv
    -rm -rf results
```

Рис. 19: Внесение изменений в Makefile

Построение графика по проведенному эксперименту

```
mininet@mininet-vm:~/work/lab_iperf3/iperf3$ make
make python lab_iperf3.py
*** Adding controller
*** Adding hosts
*** Adding switch
*** Creating links
(100.00Mbit 75ms delay) (100.00Mbit 75ms delay) (100.00Mbit 75ms delay) (100.00Mbit 75ms delay) *** Starting network
*** Configuring hosts
h1 (cfe -i/100000us) h2 (cfe -i/100000us)
*** Starting controller
ok
*** Starting 1 switches
s3 (100.00Mbit 75ms delay) (100.00Mbit 75ms delay) ... (100.00Mbit 75ms delay) (100.00Mbit 75ms delay)
*** Waiting for switches to connect
ok
Host h1 has IP address 10.0.0.1 and MAC address fa:0f:df:5c:cf:b0
Host h2 has IP address 10.0.0.2 and MAC address 1a:93:20:81:35:93
*** Starting network
*** Traffic generation
*** h2 : ('iperf3 -s -D -i',)
*** h1 : ('iperf3 -c', '10.0.0.2', '-j > iperf_result.json')
*** Stopping network*** Stopping 1 controllers
ok
*** Stopping 2 links
..
*** Stopping 1 switches
ok
*** Stopping 2 hosts
h1 h2
*** Done
plot_iperf.sh iperf_result.json
mininet@mininet-vm:~/work/lab_iperf3/iperf3$
```

Активация Windows
Чтобы активировать Windows, перейдите на [www.microsoft.com/windows/activation](#).

Рис. 20: Успешная отработка Makefile

Выводы

В результате выполнения данной лабораторной работы я познакомилась с инструментом для измерения пропускной способности сети в режиме реального времени — iPerf3, а также получила навыки проведения воспроизводимого эксперимента по измерению пропускной способности моделируемой сети в среде Mininet.